

COGS 189 Project Proposal and Group Contract

Bayesian Neural Shifts

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Project Information

Course: COGS 189

Project Title: Bayesian Neural Shifts

Group Number: Bayesian Waves

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Project Topic

This project studies **when the statistical behavior of EEG signals changes** during visual processing. We use **Bayesian Online Change-Point Detection (BOCPD)** to detect regime shifts in EEG activity in two complementary settings:

1. **Across repeated presentations of the same visual stimulus**, and
2. **Within a single stimulus-locked EEG epoch**.

Our work uses the **THINGS-EEG** dataset, a large-scale EEG benchmark for visual object recognition. The broader goal is to understand whether neural responses remain statistically stable or shift over time, and whether BOCPD can reveal meaningful transitions tied to neural adaptation, temporal processing stages, or variance changes in brain activity.

Motivation

EEG is fundamentally nonstationary. Neural responses can change within milliseconds after a stimulus appears, and they can also shift across repeated exposures to the same image due to adaptation, habituation, attention changes, or recording variability. Many standard EEG modeling approaches assume local or global stationarity, which can obscure these transitions.

Our motivation is to build a principled probabilistic framework for detecting *when* these shifts occur. Rather than only predicting EEG responses, we want to better characterize the temporal structure of neural activity. This is relevant for computational neuroscience, adaptive brain-computer interfaces, and robust modeling pipelines that need to account for distributional change.

Research Question

Primary Question: When does the statistical behavior of EEG change?

More specifically, we ask:

- Do EEG-derived features remain stationary across repeated presentations of the same image?
- Are there identifiable changepoints within a single 1-second EEG epoch after stimulus onset?
- Which features are most informative for detecting regime shifts: mean-based or variance-based features?

Objectives

The primary objectives of the project are:

1. Build an end-to-end EEG analysis pipeline for BOCPD on a large public dataset.
2. Extract interpretable scalar EEG features suitable for Bayesian changepoint detection.
3. Evaluate changepoint behavior across repetitions and within epochs.
4. Compare feature choices, especially log bandpower, windowed mean amplitude, and windowed log variance.
5. Validate the BOCPD pipeline on synthetic sequences with known changepoints.
6. Present the project through a written report, code repository, and project website.

Dataset and Resources

Resource	Link / Description
Dataset	THINGS-EEG dataset: https://osf.io/3jk45/
Project Website	https://bayesian-detection-for-eeeg.vercel.app
Code Repository	https://github.com/SatrunsDream/Bayesian-Detection-for-EEG
Primary Paper / Dataset Reference	Gifford et al. (2022), <i>A large and rich EEG dataset for modeling human visual object recognition</i>
Methods Basis	Adams and MacKay (2007), <i>Bayesian Online Changepoint Detection</i>

Table 1: Main datasets and project resources.

The THINGS-EEG dataset contains EEG recordings from 10 participants viewing natural object images in a rapid serial visual presentation paradigm. For our project, the relevant structure is:

Dataset Summary

The data is structured as a four-dimensional array:

(conditions, repetitions, channels, timepoints)

This makes it possible to study both temporal dynamics within trials and stability or drift across repeated trials.

Property	Training Partition	Test Partition
Conditions (unique images)	16,540	200
Repetitions per condition	4	80
Channels	17 posterior electrodes	17 posterior electrodes
Timepoints per epoch	100	100
Sampling rate	100 Hz	100 Hz
Time window	-200 ms to +795 ms	-200 ms to +795 ms

Table 2: Summary of the THINGS-EEG data used in the project.

Methodology

Our project uses a probabilistic changepoint detection framework built around BOCPD. The pipeline consists of the following stages:

1. Exploratory Data Analysis

We first perform EDA to verify data quality and characterize major patterns in the EEG:

- Grand-average ERP inspection
- Participant-level quality assessment
- Variance structure across time within epochs
- Spectral bandpower checks and Gaussianity assessment

2. Feature Extraction

We extract three scalar feature types:

1. **Log bandpower** for across-repetition analysis
2. **Windowed mean amplitude** for within-epoch analysis
3. **Windowed log variance** for within-epoch analysis

3. Bayesian Online Change-Point Detection

We apply BOCPD using a conjugate Gaussian observation model to estimate posterior changepoint probabilities. The analyses are divided into:

- **Across-repetition analysis:** detect whether repeated presentations of the same image show shifts in EEG feature distributions
- **Within-epoch analysis:** detect changes in the statistical regime during the 1-second post-stimulus EEG response
- **Synthetic validation:** evaluate BOCPD behavior on simulated sequences with known change-points

4. Evaluation

We evaluate results using:

- Maximum changepoint posterior probability
- Peak changepoint latency
- Cross-participant consistency
- Synthetic detection rate, latency, and false-positive rate

Expected Findings

Based on our current analyses, we expect the following:

- Some conditions will show nonstationarity across repeated presentations
- Variance-based features will be more informative than mean-based features for within-epoch changepoint detection
- BOCPD will detect meaningful transitions in the post-stimulus period, especially in later windows where variance increases

Deliverables

Our final project deliverables will include:

1. Final written report
2. Clean and reproducible code repository
3. Project website summarizing methods and findings
4. Visualizations and tables summarizing major results

Work Distribution

Member	Primary Responsibilities
Sardor Sobirov	Methodology lead; BOCPD implementation; feature design; project website; half of exploratory data analysis; integration of results into the overall technical pipeline.
Diego Arevalo Fernandez	Report writing lead; interpretation and write-up support; half of exploratory data analysis; editing and polishing final written deliverables.

Table 3: Primary division of labor for the project.

Shared responsibilities:

- Final review of results and interpretations
- Discussion of related literature
- Presentation planning and final submission checks

Group Contract

Communication

We will communicate primarily through direct messages and a shared group chat. Important project updates, task assignments, deadlines, and questions should be communicated there so both members remain aligned.

Meetings

We will meet regularly each week, either virtually or in person, to review progress, identify blockers, and assign next steps. Additional check-ins will happen as needed near major deadlines.

Expectations

Each group member agrees to:

- Complete assigned work on time
- Communicate early if deadlines or responsibilities become difficult to meet
- Review shared materials before meetings
- Contribute constructively to writing, interpretation, and final presentation quality
- Maintain organized project files, code, and documentation

Conflict Resolution

If issues arise regarding workload, communication, or missed deadlines, we will first discuss them directly as a group and attempt to rebalance tasks fairly. If needed, we will escalate unresolved issues to the instructor or teaching staff early rather than waiting until the final deadline.

Accountability

Both members are responsible for keeping the project moving forward. If one member becomes unavailable temporarily, the other may complete urgent tasks needed to keep the project on track, but workload adjustments should be acknowledged and discussed openly.